

Background

Hypertension Burden:

- Hypertension affects approximately 47% of adults in the United States and over one billion individuals worldwide.
- It is a major risk factor for cardiovascular disease, stroke, and chronic kidney disease.
- It is often asymptomatic and is commonly referred to as a "silent killer," as it can go undiagnosed while significantly increasing the risk of severe cardiovascular outcomes.

Smoking as a Risk Factor:

- Cigarette smoking is a well-established risk factor for cardiovascular disease.
- Smoking is associated with increased risk of heart attack and stroke and can also contribute to the development of hypertension.

Public Health Significance:

- Hypertension contributes to mortality; therefore, effective management of hypertension requires long-term lifestyle modifications and, in most cases, pharmacologic treatment.
- Identifying the independent contribution of smoking to hypertension risk can strengthen smoking cessation efforts to reduce the impact of cardiovascular conditions.

Purpose

Primary Objective:

- This study will evaluate the association between smoking status and hypertension by examining differences in hypertension risk across smoking categories (never, former, and current).
- The strength and direction of this association will be illustrated using statistical modeling.

Secondary Objective:

- This study examines how the association between smoking and hypertension changes after adjusting for confounders, including age and sex.
- The variations in the relationship between smoking and hypertension across BMI levels will be highlighted through interaction modeling.

Methods

Study Design:

- Cross-sectional study using NHANES 2021–2023 data (CDC)
- Participants were recruited through Mobile Examination Centers
- Analytic sample: n = 6,052 (complete cases)

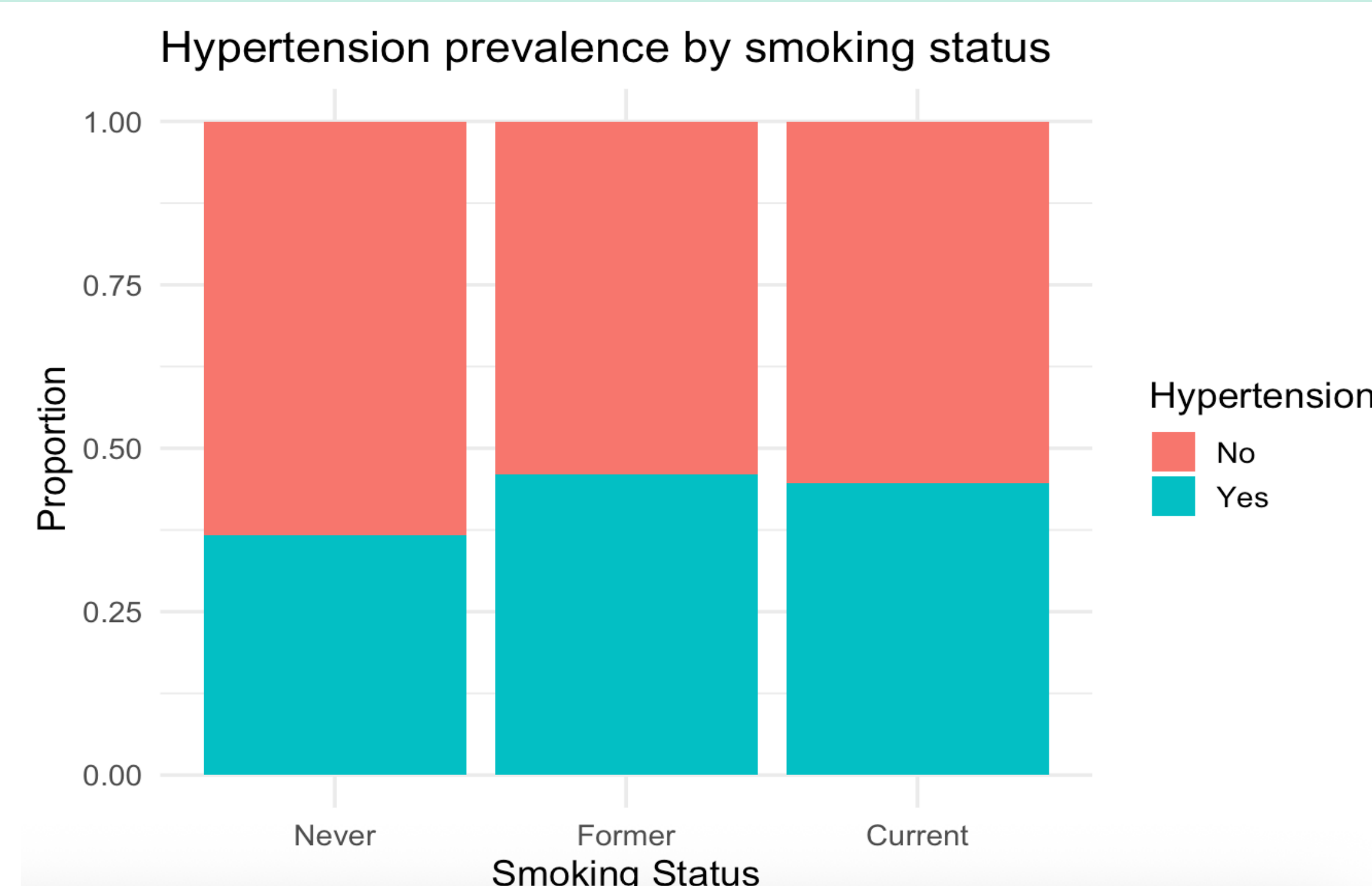
Variables:

- Outcome:** hypertension (SBP ≥ 130 mmHg or DBP ≥ 80 mmHg, mean of 3 readings)
- Predictor:** smoking status (never, former, current)
- Covariates:** age, sex, BMI

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(\text{smoking}) + \beta_2(\text{BMI}) + \beta_3(\text{smoking} \times \text{BMI}) + \beta_4(\text{age}) + \beta_5(\text{sex})$$

Model: To examine the association between smoking status and hypertension, we first fit a crude logistic regression model with age, sex, and BMI, then extend it to include smoking status and a BMI x smoking interaction term.

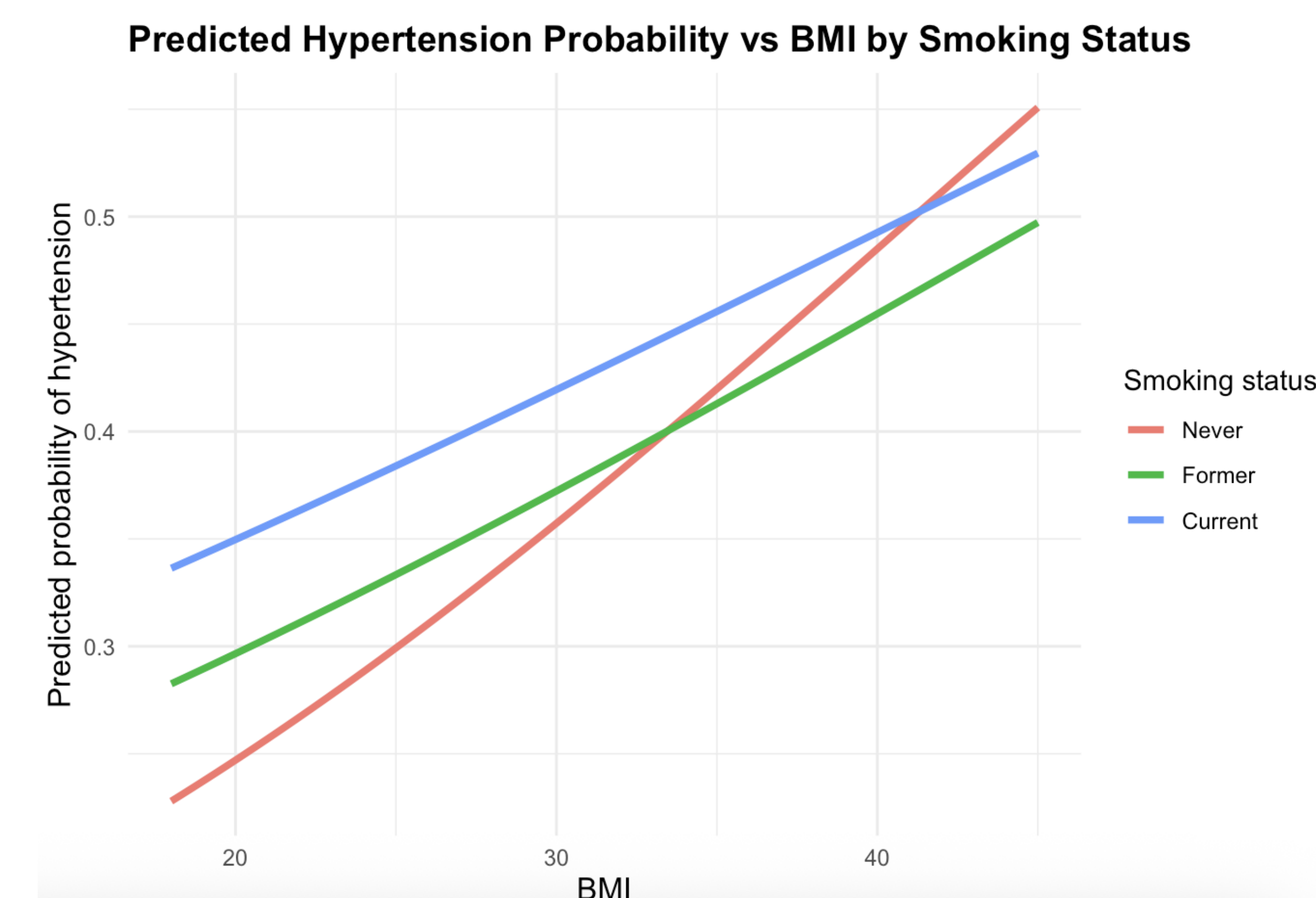
Results



Takeaway: Former and current smokers show higher hypertension prevalence compared to never smokers.

Variable	Estimate	OR	95% CI
BMI	0.045	1.05	(1.038, 1.054)
Age	0.031	1.03	(1.028, 1.035)
Sex	-0.234	0.79	(0.709, 0.882)
Smoking Former	0.058	1.06	(0.932, 1.206)
Smoking Current	0.276	1.32	(1.127, 1.54)

Results



Takeaway: At lower BMI levels, current smokers show the highest predicted probability of hypertension. However, as BMI increases, the risk of hypertension rises sharply among never-smokers. At the highest BMI levels, hypertension risk is no longer lowest among never-smokers.

Discussion

- BMI, age, sex, and current smoking status are significant predictors of hypertension in our model ($p < 0.001$).
- Current smokers show a 32% increased risk of hypertension when compared with never smokers.**
- Interestingly, **former smokers show only a 6% increased risk of hypertension compared to never smokers**, illustrating how the negative effect of smoking on hypertension outcomes may attenuate over time after quitting.

Conclusion

- Overall, former smokers **did not** show significantly higher hypertension risk compared to never smokers, and current smokers showed higher odds of hypertension compared to never smokers.
- Smoking may influence hypertension through multiple pathways, including metabolic and cardiovascular mechanisms.
- Among the highest risk groups, traditional metabolic risk factors such as age and BMI may be more important than smoking status.